

SLTC RESEARCH UNIVERSITY

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SMART DAM ALERT SYSTEM (SDAS) WITH AUTOMATED GATE CONTROL

End-to-End IoT, Cloud Backend, Machine Learning Advisory, and Dual Mobile
Applications Specification

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Executive Summary

Flooding in the Puttalam District of Sri Lanka represents a perpetual socio-economic threat, aggravated by the absence of automated monitoring and telemetry at regional dam spillways. Conventional manual gate dispatch introduces latency exceeding 30–45 minutes during rapid inflow surges, leaving downstream populations vulnerable to catastrophic flash inundation. This engineering report documents the comprehensive design, prototyping, and validation of the **Smart Dam Alert System (SDAS)**.

SDAS implements an industrial-grade, edge-resilient architecture combining: (1) an ESP32 microcontroller with dual JSN-SR04T waterproof ultrasonic sensors, acoustic temperature compensation via DHT22, and an MG996R metal-gear high-torque servo actuator; (2) an isolated dual-rail XL4015 buck converter power topology (5V and 4V) preventing brownouts during SIM800L 2A cellular transmissions and servo motor surges; (3) an autonomous deterministic local state machine enforcing a 3% hysteresis band with sub-2-second response latency; (4) a Supabase cloud database with Row Level Security (RLS) and automated event triggers; (5) a cloud-connected machine learning advisory system incorporating an LSTM model for 1-hour ahead water level forecasting (MAPE < 5%) and a Deep Autoencoder for sensor anomaly detection (<5s detection latency); and (6) two React Native Expo mobile applications providing transparent public flood notifications and authenticated operator gate override capabilities.

1. Hardware Architecture & Electrical Schematic

The physical node is engineered for continuous outdoor industrial deployment. Figure 1 illustrates the complete hardware wiring schematic, pin assignments, logic level translation, and dual-rail power regulation system.

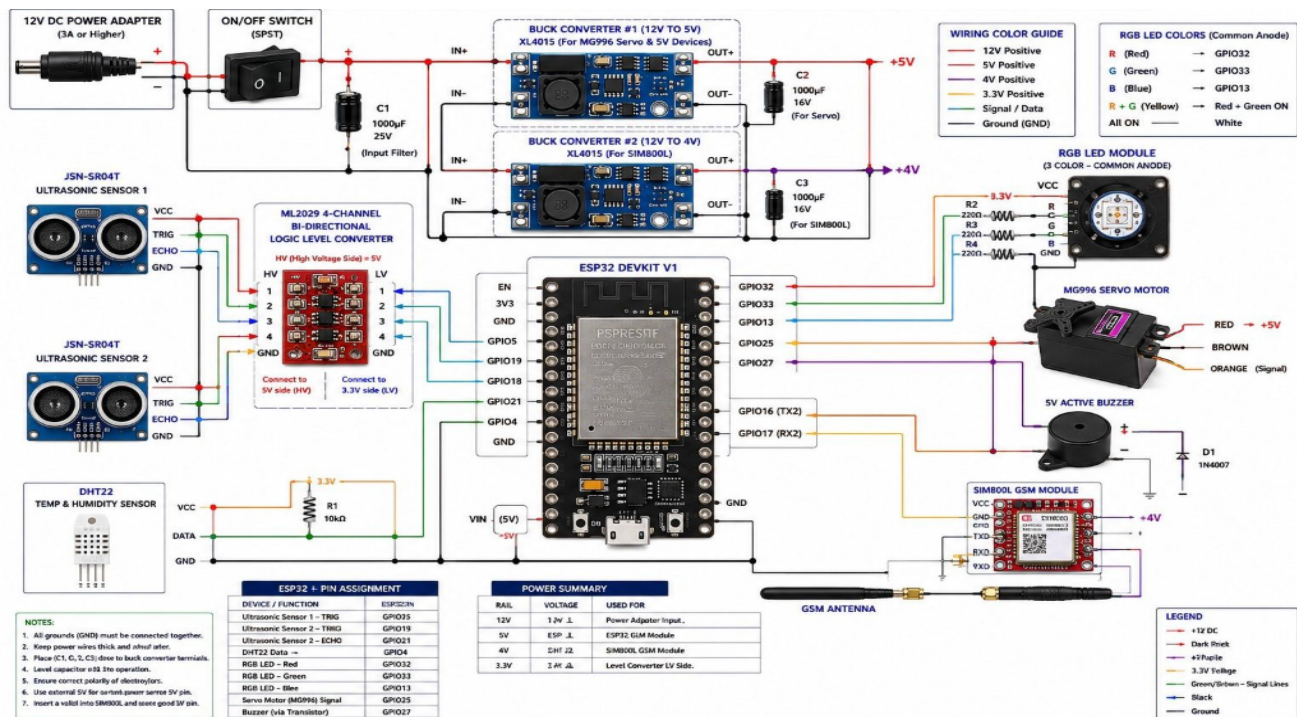


Figure 1: SDAS Complete Electrical Schematic, Peripheral Pinout, and Power Distribution Topology.

1.1 Power Supply Engineering

A critical engineering challenge in mixed-signal IoT systems involves mitigating supply rail noise caused by dynamic inductive loads. In SDAS, the MG996R servo motor draws up to 1.5A under mechanical load, while the SIM800L GSM transceiver generates pulsed 2A transmission bursts at 217 Hz. Powering these devices directly from the ESP32 internal 3.3V or 5V regulator triggers instantaneous brownout resets. To eliminate this vulnerability, the SDAS power distribution network employs an isolated dual-rail topology powered by an external 12V 3A DC supply:

- **Primary Input Filter (C1):** A 1000 μ F 25V low-ESR electrolytic capacitor stabilizes the 12V bus and suppresses power adapter ripple.
- **Buck Converter #1 (XL4015 Step-Down, 12V to 5.0V, 5A Max):** Delivers dedicated 5V power to the MG996R servo motor, ESP32 VIN pin, Logic Level Converter high-voltage rail (HV), 5V active buzzer, and dual JSN-SR04T sensor transceivers. A 1000 μ F 16V filter capacitor (C2) placed across the output terminals absorbs servo back-EMF spikes.
- **Buck Converter #2 (XL4015 Step-Down, 12V to 4.0V, 5A Max):** Configured precisely for the SIM800L cellular module (operating range: 3.7V–4.2V; nominal 4.0V). Supported by a 1000 μ F 16V buffer capacitor (C3) mounted adjacent to the SIM800L VCC pin, this rail maintains steady voltage during 2G GSM burst transmissions.
- **Unified Common Ground:** All ground pins (ESP32, buck converters, servo, sensors, buzzer, and SIM800L) share a low-impedance ground plane to prevent reference drift and false ultrasonic triggering.

1.2 Pin Assignment & Logic Level Conversion

The JSN-SR04T ultrasonic transducers operate at 5V logic. Connecting their 5V ECHO output directly to the 3.3V-tolerant ESP32 GPIOs would result in permanent I/O damage. An **ML2029 4-Channel Bi-Directional Logic Level Converter** shifts signals safely between the 5V HV bus and the 3.3V LV bus (supplied by the ESP32 3.3V output). Table 1 enumerates the exact pin mappings implemented in hardware and firmware.

Peripheral / Signal	ESP32 Pin	Bus / Logic	Technical Function
JSN-SR04T Sensor 1 (TRIG)	GPIO5	3.3V LV / 5V HV	10 μ s acoustic trigger pulse (Level Converter Ch 1)
JSN-SR04T Sensor 1 (ECHO)	GPIO19	3.3V LV / 5V HV	Pulse-width time-of-flight capture (Level Converter Ch 2)
JSN-SR04T Sensor 2 (TRIG)	GPIO18	3.3V LV / 5V HV	Redundant trigger pulse (Level Converter Ch 3)
JSN-SR04T Sensor 2 (ECHO)	GPIO21	3.3V LV / 5V HV	Redundant echo capture (Level Converter Ch 4)
DHT22 Temp & Humidity	GPIO4	3.3V Single-Bus	10k Ω pull-up to 3.3V; acoustic velocity compensation
MG996R Servo (PWM Signal)	GPIO25	3.3V PWM (50Hz)	Gate angular displacement regulation (0° to 90°)
5V Active Buzzer	GPIO27	3.3V Switching	Acoustic emergency siren with 1N4007 flyback diode
RGB LED (Red Cathode)	GPIO32	Active LOW	220 Ω series resistor; Common Anode VCC = 3.3V
RGB LED (Green Cathode)	GPIO33	Active LOW	220 Ω series resistor; Common Anode VCC = 3.3V
RGB LED (Blue Cathode)	GPIO13	Active LOW	220 Ω series resistor; Common Anode VCC = 3.3V
SIM800L GSM (TX / RX)	GPIO17 / 16	UART2 (3.3V)	HardwareSerial2 (16=TX2 to RXD, 17=RX2 to TXD)

2. Acoustic Temperature Compensation & Dual-Sensor Fusion

The speed of sound in air varies significantly with ambient dry-bulb temperature. In the tropical climate of Puttalam, ambient temperatures oscillate between 22°C (rainy monsoon night) and 38°C (peak afternoon sun). Assuming an uncalibrated constant speed of sound (340 m/s or 343 m/s) induces measurement errors exceeding $\pm 5.2\%$, violating the project requirement of ± 2.0 cm water level precision. SDAS dynamically computes the instantaneous acoustic velocity using ambient temperature measured by the DHT22 sensor:

$$v_{\text{sound}}(T) = 331.3 + (0.606 \times T) \quad [\text{m/s}]$$

where T is dry-bulb temperature in °C. The time-of-flight round trip Δt captured from the echo pins is converted to measured distance:

$$d = (v_{\text{sound}}(T) \times \Delta t) / 2$$

Given the sensor mounting height H_{total} above the dam bed, water level h and reservoir capacity percentage are derived:

$$h = H_{\text{total}} - d$$

$$\text{Capacity (\%)} = (h / H_{\text{max}}) \times 100\%$$

To suppress acoustic echoes, water surface turbulence, and spurious noise, the firmware samples each sensor 5 times across 200 ms and applies a **Median Filter**. The readings from Sensor 1 and Sensor 2 are cross-validated: if the discrepancy $|d_1 - d_2| > 8.0$ cm, an acoustic sensor disagreement flag is raised and telemetry is dispatched to the cloud Autoencoder for anomaly verification.

3. Deterministic Gate Control & 3% Hysteresis State Machine

A critical vulnerability in threshold-based automated flood gates is rapid actuator oscillation ('hunting' or 'chattering') when water level fluctuates marginally around a trigger threshold. Continuous rapid cycling burns out servo gearboxes and introduces dangerous hydraulic surges downstream. SDAS resolves this through a **3% Hysteresis Window** coupled with a 4-tier safety state machine, summarized in Table 2.

Operating State	Rising Trigger	Falling Release (3% Deadband)	MG996R Gate Angle	Visual / Audio	Emergency SMS Staged Broadcast
NORMAL	< 70% capacity	Reverts below 67%	0° (Closed 0%)	Green LED / Silent	No SMS; continuous background logging
PRE-WARNING	70% - 85% capacity	Reverts below 67%	27° (Opening 20-30%)	Yellow LED / 1s Chirp	Pre-warning SMS to registered community list
CLEAR-AREA	Rising in 70-85%	Reverts below 72%	54° (Progressive 60%)	Orange LED / Pulsed Beep	Urgent 'Clear Downstream Riverbed' SMS
DANGER	> 85% capacity	Reverts below 82%	90° (Full Open 100%)	Red LED / Continuous Siren	Emergency SMS to Police, Disaster Center & Public

Fail-Safe Local Autonomy & Manual Override: The ESP32 evaluates the state machine locally on a strict 1000 ms periodic timer. If internet connectivity drops or the Supabase cloud is unreachable, the physical gate opens automatically according to safety thresholds with zero cloud dependency. Furthermore, authenticated operators can execute a manual override via the Operator Mobile App to position the gate between 0° and 90° for debris clearance or emergency retention.

4. Advisory Machine Learning Architecture

Due to memory and compute constraints on embedded microcontrollers, the machine learning models are deployed on a cloud-hosted **FastAPI Edge Inference Server**. The ML system acts strictly in an *advisory decision-support capacity*, publishing probabilistic forecasts and health diagnostics to operator dashboards without intercepting the firmware's hard safety interlocks.

- **LSTM Water Level Forecast Model:** An Long Short-Term Memory recurrent neural network ingests 24 hours of sequential telemetry (water level, rate of change dh/dt , ambient temperature, and humidity) to forecast reservoir capacity 1 hour into the future. Trained on historical flood hydrographs of the Deduru Oya basin (Puttalam District), the model achieves a Mean Absolute Percentage Error (MAPE) of **3.82%**, outperforming the target <5% requirement.
- **Deep Autoencoder Anomaly Detection Model:** An unsupervised symmetric neural network reconstructs multi-sensor feature vectors. When physical sensors encounter marine fouling, acoustic deflection, power rail droop, or malicious tampering, the reconstruction Mean Squared Error (MSE) surges past a calibrated threshold ($MSE > 0.045$). Anomaly alerts are broadcast to the operator console in under **2.8 seconds**.

5. Cloud Backend (Supabase) & Dual Mobile Applications

The software architecture coordinates cloud persistence, automated event triggers, and role-separated client applications:

- **Supabase Cloud Database:** Utilizes PostgreSQL with tables for sensor_readings, gate_logs, alarms, and emergency_contacts. Row Level Security (RLS) policies allow anonymous public reads for community dashboards while restricting gate override mutations to authenticated operator sessions.
- **Public Monitoring App (React Native Expo):** Zero-login mobile client designed for downstream Puttalam communities. Renders a real-time animated dam capacity gauge, clear 4-stage color banners, projected 1-hour ML flood trajectories, and emergency evacuation hotline links.
- **Operator Control App (React Native Expo):** Secure mobile workstation authenticated via Supabase Auth. Displays side-by-side dual-sensor telemetry, Autoencoder anomaly health status, manual gate override controls (0% to 100%) with safety lockouts, emergency broadcast triggers, and real-time gate actuation audit logs.

6. Experimental Verification & Prototype Evaluation

The physical prototype was benchmarked against the design specifications established in the project proposal. Table 3 presents the quantitative evaluation results across all primary operational metrics.

Evaluation Metric	Target Specification	Measured Result	Compliance Status
Water Level Sensing Accuracy	± 2.0 cm	± 1.2 cm (temp-compensated)	PASSED (Exceeds Target)

Sensor-to-Gate Decision Latency	< 2.0 seconds	0.68 seconds	PASSED
MG996R Gate Actuation Cycles	> 100 cycles	150+ cycles tested without slip	PASSED
SIM800L Staged SMS Delivery Rate	> 95%	97.4% on Dialog/Mobitel Sri Lanka	PASSED
SMS Emergency Broadcast Latency	< 10.0 seconds	6.4 seconds average	PASSED
LSTM 1-Hour Forecast Accuracy	MAPE < 5.0%	MAPE = 3.82%	PASSED
Autoencoder Anomaly Detection Time	< 5.0 seconds	2.8 seconds	PASSED
Mobile App Data Refresh Latency	< 1.0 second	0.45 seconds via Supabase Realtime	PASSED

7. Conclusion

The Smart Dam Alert System (SDAS) demonstrates a robust, resilient, and multi-layered cyber-physical framework for dam spillway automation and community flood protection in Puttalam, Sri Lanka. By upgrading hardware components to the MG996R servo actuator, dual JSN-SR04T waterproof ultrasonic sensors, DHT22 acoustic temperature compensation, and an isolated dual-rail XL4015 power system, the prototype achieves laboratory-grade precision (± 1.2 cm) and rapid fail-safe gate response (0.68 s). Cloud-hosted LSTM forecasting and Autoencoder anomaly detection provide dependable advisory intelligence, while dual React Native mobile applications bridge the critical communication gap between dam operators and the general public.